

CHEM 515 Practice Session 5: Correlated wavefunction methods; energy decomposition analysis for non-covalent interactions

Name: _____

In this practice, you will first practice (i) using a combination of correlated wavefunction methods to generate high-level quantum chemistry reference data, and (ii) calculating singlet-triplet gap of the Be atom using CASSCF and related methods. You will then employ the ALMO-EDA suite of methods to investigate (i) the $\text{H}_2\text{O} \cdots \text{Cl}^-$ interaction in the gas phase and in implicit solvent; (ii) vibrational frequency shift of CO in the $\text{OBe} \cdots \text{CO}$ complex.

You should use the Ictotck server to complete **Exercises 1 & 3**. To avoid over-burdening the server, please run Q-Chem jobs with “`qchem -nt 4 [input] [output] &`”. The input files can still be generated using IQmol, but if you are getting more proficient with text editing on Linux and are relatively familiar with the commonly used keywords in Q-Chem input files, you may find it much faster to make copies and modify the text files directly on the server, especially for jobs that only differ by one or two keywords.

Exercise 2 involves a few quick ORCA calculations. You will should complete them on SDSC Expanse so that you can practice the usage of the “`submit_orca`” script. **Exercise 4** involves only one Q-Chem job, which will take some time to finish. You are supposed to submit that to the “shared” queue on Expanse to complete the calculation (see detailed instructions below).

You will need to **complete the tables and short-answer questions on this handout** and also submit the files that are requested. Please refer to the **Assignment page** on Canvas for detailed submission guidelines. For this practice, **9 points** are associated with the completion of this handout and only **1 point** is for attendance.

Exercise 1: Generating the quantum chemistry reference value for the $\text{HCN} \rightarrow \text{HNC}$ isomerization energy

In this exercise, we will calculate the energy change upon the $\text{HCN} \rightarrow \text{HNC}$ isomerization reaction at the MP2/CBS and $\Delta\text{CCSD(T)}/\text{CBS}$ levels. The isomerization energy of HCN is defined as

$$\Delta E_{\text{iso}} = E(\text{HNC}) - E(\text{HCN}) \quad (1)$$

The MP2/CBS energy in this example is defined as the sum of HF/aug-cc-pV5Z energy and MP2 correlation energy extrapolated to the CBS limit, i.e.,

$$\begin{aligned} E_{\text{MP2}}(\text{CBS}) &\approx E_{\text{HF}}(\text{a5Z}) + E_{\text{MP2}}^{\text{corr}}(\text{CBS}) \\ &\approx E_{\text{HF}}(\text{a5Z}) + [X^3 E_{\text{MP2}}^{\text{corr}}(X) - Y^3 E_{\text{MP2}}^{\text{corr}}(Y)] / (X^3 - Y^3) \end{aligned} \quad (2)$$

Here we use RI-MP2 results calculated using aug-cc-pVTZ and aug-cc-pVQZ to conduct the basis set extrapolation, for which X and Y will be 3 and 4, respectively. Let's start with RI-MP2 calculations for the given `HCN.xyz` structure. Note that for this exercise, the geometries `HCN.xyz` and `HNC.xyz` are taken from the BH76 subset in GMTKN55, and thus **no further geometry optimization should be performed**. To set up an RIMP2 calculation using IQmol:

- Set Method to “RI-MP2” and Basis to “aug-cc-pVTZ” or “aug-cc-pVQZ”
- Set the “RI Fitting Basis” at the bottom to “RIMP2-aug-cc-pVTZ” or “RIMP2-aug-cc-pVQZ”
- Set GUI = 0 and the SCF convergence criterion to 8

Through the output of an RI-MP2 calculation, you will be able to find:

- The Hartree-Fock (HF) energy (the number after “SCF energy”)
- The RI-MP2 total energy (“RIMP2 total energy”)
- The RI-MP2 correlation energy (“RIMP2 correlation energy”)

The section showing these numbers can be easily seen by searching for “RIMP2 correlation energy” in the output.

You can now report these values [HF, RIMP2, and RIMP2(corr)] based on your RIMP2/aug-cc-pVTZ and aug-cc-pVQZ calculations in **Table 1**. In addition, you need to run a Hartree-Fock calculation with the aug-cc-pV5Z basis (remember to set SCF_CONVERGENCE to 8) and fill in the blank “HF/a5Z”. **(0.6 pt)**

Now use Eq. (2) to obtain the RIMP2/CBS result for HCN. Note that the extrapolation should only be applied to the RIMP2 correlation energy rather than the RIMP2 total energy. You should use Excel to do the extrapolation so that you can use the same equation later for HNC. **(0.2 pt)**

Table 1: Worksheet to obtain the MP2/CBS and $\Delta\text{CCSD(T)}/\text{CBS}$ values for the $\text{HCN} \rightarrow \text{HNC}$ isomerization energy. aTZ, aQZ, and a5Z stand for aug-cc-pVTZ, aug-cc-pVQZ, and aug-cc-pV5Z, respectively. Keep at least **6 decimals** for numbers in a.u., and use **2 decimals** for numbers in the rightmost column (1 a.u. = 627.5095 kcal/mol).

Item	$E(\text{HCN})$ (a.u.)	$E(\text{HNC})$ (a.u.)	ΔE_{iso} (kcal/mol)
HF/aTZ			
HF/aQZ			
HF/a5Z			
RIMP2/aTZ			
RIMP2/aQZ			
RIMP2(corr)/aTZ			—
RIMP2(corr)/aQZ			—
RIMP2/CBS			
CCSD/aTZ			
CCSD(T)/aTZ			
$\Delta\text{CCSD(T)}/\text{CBS}$			

Now repeat the procedure above for the HNC molecule (using the given geometry “HNC.xyz”) and report the ΔE_{iso} values. **(1 pt)** Up to this point, you should have completed **Table 1** up to the row “RIMP2/CBS”.

The $\Delta\text{CCSD(T)}/\text{CBS}$ values in our case can be calculated using the formula below:

$$E_{\Delta\text{CCSD(T)}}(\text{CBS}) = E_{\text{MP2}}(\text{CBS}) + [E_{\text{CCSD(T)}}(\text{aTZ}) - E_{\text{MP2}}(\text{aTZ})] \quad (3)$$

Since we have already done the RIMP2/aTZ calculation, we only need to perform a CCSD(T) calculation with the aug-cc-pVTZ basis for each of the species. You can use IQmol to set up this calculation (setting Method to “CCSD(T)”); also remember to set GUI = 0 and SCF_CONVERGENCE = 8). Search for “CCSD total

energy” and “CCSD(T) total energy” in the output, report the numbers for both HCN and HNC in **Table 1**, and then calculate the corresponding ΔE_{iso} values. **(0.4 pt)**

You can now use Eq. (3) to calculate the $\Delta\text{CCSD(T)}/\text{CBS}$ energies for both species. Perform the calculations on Excel and report the numbers (including the final ΔE_{iso} value) in **Table 1**. **(0.2 pt)**

The ΔE_{iso} value evaluated at the $\Delta\text{CCSD(T)}/\text{CBS}$ level can be regarded as the *quantum chemistry reference*. With the reference value we can assess the performance of DFT functionals. Now calculate the isomerization energy using the B97M-V and $\omega\text{B97M-V}$ functionals the def2-QZVPP basis. Report your results in **Table 2** below. **(0.4 pt)**

Table 2: The HCN $\rightarrow$ HNC isomerization energy calculated using B97M-V and $\omega\text{B97M-V}$ with the def2-QZVPP basis set

Functional	$E(\text{HCN})$ (a.u.)	$E(\text{HNC})$ (a.u.)	ΔE_{iso} (kcal/mol)
B97M-V			
$\omega\text{B97M-V}$			

Now answer the following questions:

1. Comment on the accuracy of RIMP2/CBS and CCSD/aTZ for the prediction of the isomerization energy (e.g., overestimate or underestimate? by how much?). In terms of the absolute error, do the two density functionals in **Table 2** perform better or worse than RIMP2/CBS? **(0.4 pt)**
2. Which functional gives a smaller error compared to the $\Delta\text{CCSD(T)}/\text{CBS}$ reference? Does this align with your expectation based on the Jacob’s ladder? **(0.2 pt)**

Exercise 2: Calculating the singlet-triplet gap of the Be atom

In Lecture 16, we mentioned that while the electronic configuration of a Be atom can usually be written as $1s^2 2s^2$, its ground state (singlet) possesses some multireference character due to the close-lying $2s$ and $2p$ orbitals. In this Exercise, we will use Hartree-Fock (HF) theory and the complete active-space self-consistent field (CASSCF) method to determine the energy difference between the lowest-lying singlet and triplet states of Be, which is also known as the singlet-triplet (S-T) gap.

You are supposed to complete this exercise using the ORCA 5.0.1 available on **SDSC Expanse**. You should already have access to the “`submit_orca`” script. With that, you can submit the calculations involved in this exercise simply by “`submit_orca xxx.inp`”. No “-p” option is needed because these jobs are minimal and you don’t need to run these jobs with the OpenMPI parallelization.

We will first calculate the S-T gap using the HF theory. As a starting template, an ORCA input file “`Be_singlet_RHF.inp`” has been provided, which is for a restricted HF (RHF) calculation for the singlet state using the def2-TZVPP basis. Take a look at the content of the input file, and then run this job on the cluster. The RHF energy is reported after “**FINAL SINGLE POINT ENERGY**” in the output file (same for the other methods below unless otherwise noted). From the “**ORBITAL ENERGIES**” and “**LOEWDIN REDUCED**

ORBITAL CHARGES” sections in the output, you will be able to figure out that this corresponds to the $1s^2 2s^2$ configuration.

Then we are going to calculate the triplet state energy using unrestricted Hartree-Fock (UHF). To do that, you simply need to make a copy of the RHF input and change the multiplicity (the 2nd number after “xyz” in the input) from “1” to “3”. Run the UHF calculation, and report the RHF/UHF energies for the singlet/triplet states in **Table 3** below, and then calculate the S-T gap at the HF level. **(0.3 pt)**

Table 3: Worksheet to determine the singlet-triplet gap (ΔE_{S-T}) of a Be atom (1 a.u. = 27.2114 eV)

Method	Singlet (a.u.)	Triplet (a.u.)	ΔE_{S-T} (eV)	Rel. Error (%)
RHF/UHF				
CASSCF(2,4)				
SA-CASSCF(2,4)				
NEVPT2				

The Hartree-Fock theory is unable to describe systems of a multireference character because the HF wavefunction is represented as a single Slater determinant. To properly capture the multireference nature of the Be atom, we are now using CASSCF to calculate the energies of the lowest singlet and triplet states of Be. The Be atom has two valence electrons and four valence orbitals ($2s$, $2p_x$, $2p_y$, $2p_z$) that are close in energy. Therefore, we will place 2 electrons in an active space constructed by 4 orbitals in our CASSCF calculations, which is denoted as CASSCF(2,4).

To set up the CASSCF(2,4) calculations, first make a copy of the original input “Be_singlet_RHF.inp” and rename it as “Be_singlet_CASSCF.inp”. Then you need to add the following section to the new input, between the first line and the line “* xyz 0 1”:

```
%casscf
nel 2
norb 4
end
```

Create the CASSCF input for the triplet state in the same way. Then you can submit both jobs to the cluster using “submit_orca Be*CASSCF.inp”, if your input files are named properly. After the two jobs finish, report the CASSCF energies (after “FINAL SINGLE POINT ENERGY”) in **Table 3**, and then calculate the value of ΔE_{S-T} at the CASSCF(2,4) level. **(0.3 pt)**

Above we performed two separate CASSCF calculations for the singlet and triplet states. It is also possible to perform a single CASSCF calculation that gives both singlet and triplet energies. This approach is called the **state-averaged (SA)-CASSCF**. Within SA-CASSCF, the singlet and triplet states share the same set of orbitals, and the orbitals are optimized by minimizing the average of the singlet and triplet energies, i.e., $(E_{\text{sing}} + E_{\text{trip}})/2$. To set up an SA-CASSCF calculation for our problem, first make a copy of “Be_singlet_CASSCF.inp” and rename it as “Be_SA_CASSCF.inp”, then add two extra lines under the %casscf section (right below “norb 4”):

```
mult 1,3
nroots 1,1
```

This tells the program to do state-averaging with one singlet state and one triplet state. After completing the SA-CASSCF calculation, search for “CAS-SCF STATES FOR BLOCK” in the output file. You should be able to see two blocks, one with “MULT= 3” (triplet) and another with “MULT= 1” (singlet). The energy value printed under each block is what you need for the “SA-CASSCF” row in **Table 3**. Then you can work out the S-T gap given by this method. **(0.2 pt)**

Finally, we are going to assess the impact of dynamic electron correlation on the S-T gap. To incorporate dynamic correlation, here we will perform a NEVPT2 (which stands for N-electron valence state 2nd-order

perturbation theory) calculation on top of the SA-CASSCF calculation. To set up this, first create a copy of “Be_SA_CASSCF.inp” and rename it as “Be_SA_CASSCF_NEVPT2.inp”. Then within the input file, you just need to add another line under the %casscf section:

```
ptmethod sc_nevpt2
```

Here “SC” stands for “strongly contracted”, which is just one flavor of the NEVPT2 method. Submit this calculation to the cluster. Once it finishes, search for section “NEVPT2 TOTAL ENERGIES” in the output file, under which you will find the energies for the triplet and singlet states in a.u. Report them in **Table 3** and then calculate the S-T gap at the SA-CASSCF(2,4)+NEVPT2 level of theory. **(0.2 pt)**

Complete the following:

1. A reference value for the S-T gap of Be atom is $21,990\text{ cm}^{-1}$ (**2.727 eV**). Based on this, calculate the relative error (in %) of each method and report them in the rightmost column of **Table 3**. Which method gives the best agreement with the reference value? **(0.25 pt)**
2. The HF theory misses both dynamic and nondynamic electron correlation. Based on the results we have obtained, which one is the main culprit for its inaccurate prediction of Be’s S-T gap? Explain your answer. **(0.25 pt)**
3. One student argues that while the singlet state of the Be atom exhibits some multireference character, the triplet state has no multireference character and the electron configuration can be well described by $[\text{He}]2s^12p^1$. Do you agree with them? Explain your answer. **(0.3 pt)** (**Hint:** you may want to check the occupation numbers and composition of the orbitals in “Be_triplet_CASSCF.out”).

Exercise 3: Energy decomposition analysis for the $\text{H}_2\text{O}\cdots\text{Cl}^-$ complex

In this exercise, we will use the second-generation ALMO-EDA to investigate the interaction between a water molecule and a Cl^- anion in the gas phase and in implicit solvent. All calculations will be performed using the given XYZ structure file “water_cl.xyz” at the $\omega\text{B97X-V}/\text{def2-TZVPD}$ level of theory. **Please complete this exercise on the Itcotck server**, since the version of Q-Chem on SDSC Expanse is a little old (6.0.2) and does not have a new feature (non-perturbative charge-transfer analysis) that we need.

Let’s first set up the gas-phase ALMO-EDA calculation with the following steps:

- Open “Q-Chem Setup”. First change the charge of the system to -1 and make sure that the multiplicity is 1. Then you need to specify the charge and multiplicity of each fragment *manually*: fragment 1 is the chloride anion so should be “ $-1\ 1$ ”, while fragment 2 is the water molecule which should be “ $0\ 1$ ”. You also need two dashes “--” to indicate the beginning of each fragment. The completed \$molecule section should look like:

```

$molecule
-1 1
--
-1 1
Cl -1.1242249    0.0179827    0.0000000
--
0 1
O 1.9940516    -0.0909265    0.0000000
H 1.0163933    -0.2184602    0.0000000
H 2.0588108    0.8652887    0.0000000
$end

```

- Under “Calculate” select “Energy Decompose” (which gives you “JOB_TYPE = EDA”), then manually add “EDA2 = 1” which stands for “decomposition of frozen energy + FERF-DQ polarization”
- Set the functional to “wB97X-V” and basis to “def2-TZVPD”; finally, set the SCF_CONVERGENCE to 8 and set GUI = 0.

You can now run the gas-phase EDA calculation on the server. Once finished, from the output you should be able to find results for the total interaction energy (INT) and its **5 components**: permanent electrostatics (ELEC), Pauli repulsion (PAULI), dispersion (DISP), polarization (POL), and charge transfer (CT). Here we focus on the results of the “classical” version of frozen decomposition so the keywords we need to search for in the output file are:

INT: E_int ELEC: E_cls_elec PAULI: E_mod_pauli DISP: E_disp POL: E_pol CT: E_vct

Note that the energy components printed by Q-Chem are in kJ/mol. Report the numbers under the “Gas phase” column of **Table 4 (0.3 pt)**. **Then answer the following questions:**

1. Among the 5 energy components, which ones are attractive (< 0) and which ones are repulsive (> 0)? **(0.2 pt)**
2. It was proposed that hydrogen bonding interactions (like the interaction between H_2O and Cl^-) are dominated by charge-transfer interactions. Does our result support this argument? If not, which attractive force makes the largest contribution to the $\text{H}_2\text{O} \cdots \text{Cl}^-$ interaction? **(0.25 pt)**
3. Based on the ALMO-EDA results, estimate by how much (in %) the interaction energy would be underestimated if the employed density functional were unable to describe *any* dispersion interaction. **(0.25 pt)**

Let’s now set up the ALMO-EDA-II calculation with implicit solvent (SMD water). To add the solvent model, go to “Advanced → Solvent Model”, and select “SMD” as the “Solvent Method”. You may need to click that twice to make sure that “SOLVENT water” is specified under the \$smx input section. Perform this calculation on the server and complete the “SMD water” column in **Table 4** based on the output. For permanent electrostatics and Pauli repulsion, you should look for E_cls_elec(solvr) and E_mod_pauli(solvr) this time. **(0.3 pt)**

Table 4: ALMO-EDA results for the $\text{H}_2\text{O} \cdots \text{Cl}^-$ complex using $\omega\text{B97X-V/def2-TZVPD}$ (energy components in kJ/mol)

Energy components	Gas phase	SMD water
ELEC		
PAULI		
DISP		
POL		
CT		
INT		

Comparing the EDA results for $\text{H}_2\text{O} \cdots \text{Cl}^-$ in the gas phase and in the solvation environment, **discuss the effects of the solvent environment on the EDA components**. You should aim to cover the following points: (i) how much the total interaction energy changes; and (ii) which energy components are significantly impacted by the solvation environment, and by how much they change. **(0.3 pt)**

The final task of this exercise is to perform the non-perturbative polarization and charge-transfer analysis and generate the plots of complementary occupied-virtual pairs (COVPs) that describe the polarization and charge-transfer effects. Make a copy of your input for the gas-phase EDA calculation, and then make the following modifications:

- Add the following keywords to the `$rem` section:
 - `EDA_POL_A = TRUE` (turn on non-perturbative polarization analysis)
 - `EDA_VCT_A = TRUE` (turn on non-perturbative charge-transfer analysis)
 - `EDA_SAVE_COVP = TRUE` (generate plots for significant COVPs)
 - `EDA_COVP_THRESH = 2000` (only print/plot COVPs with energetic contributions > 2 kJ/mol)
 - `MAKE_CUBE_FILES = TRUE`
 - `PLOTS = TRUE`

The last two keywords are necessary for generating the plots as “`xx.cube`” files, which can be visualized using IQmol.

- Add a `$plots` section to the input to set up a plotting grid for the cube files. Here we place 80 grid points along each (X, Y, Z) direction.

```
$plots
  grid_points 80 80 80
$end
```

The calculation is ready to go. Once it is finished, complete **Table 5** based on the results of non-perturbative POL and CT analysis. The polarization energy of each fragment can be found under section “**Fragment-wise P Connection POL Analysis**”, while the decomposition of the charge-transfer term can be found under “**Fragment-wise Non-Perturbative CT Analysis**”. **(0.2 pt)**

Hint: (i) In the 2×2 matrix under “**Fragment-wise CT energies**”, the off-diagonal entries (row 1, col 2; and row 2, col 1) correspond to the charge transfer between fragments (in the row-to-column order); (ii) in your calculation, Cl^- is fragment 1, and H_2O is fragment 2.

Table 5: Energetic results (in kJ/mol) of non-perturbative polarization and charge-transfer analysis for the $\text{Cl}^- \cdots \text{H}_2\text{O}$ complex.

	POL(Cl^-)	POL(H_2O)	CT($\text{Cl}^- \rightarrow \text{H}_2\text{O}$)	CT($\text{H}_2\text{O} \rightarrow \text{Cl}^-$)
ΔE (kJ/mol)				

Complete the following:

- Based on the results in **Table 5**, which fragment contributes more to the polarization energy? Is the charge transfer primarily from Cl^- to H_2O or the opposite? **(0.2 pt)**
- The **cube** files for the significant COVPs can be found in the [jobname].plots folder. Their detailed information can be found under “**Fragment P Connection POL COVP Analysis**” and “**Non-Perturbative CT COVP Analysis**”. Within each COVP, there is a *donor* orbital and an *acceptor* orbital. The associated polarization/charge-transfer process can be viewed as electrons going from the donor orbital to the acceptor orbital. The numbers under “**idx(D)**” and “**idx(A)**” correspond to their indexes among the cube files. For instance, if “**idx(D) = 3**” and “**idx(A) = 4**” for a COVP, then the donor orbital can be visualized through “**covp_a_3.cube**”, and the acceptor orbital corresponds to “**covp_a_4.cube**”.

Plot the significant COVPs for the polarization and charge-transfer effect within this complex. A few specific requirements:

- Please use white background for IQmol; use **isovalue 0.05** for the orbital plots.
- Each pair of donor and acceptor orbitals should be plotted in the same figure. For the acceptor orbitals, use “**Lines**” instead of “**Fill**”, which can be set by double-clicking “**Cube Data**” under Model View, and then select “**Lines**”.
- Change the **Opacity** to about 50% for both the donor and acceptor orbitals
- Save the figures as **.png** files. Use a descriptive name for each figure, e.g., **ct_covp_cl_to_water.png**.

Submit the generated figures for significant COVPs with your report **(0.3 pt)**. Based on the visualization, describe the characters of the orbitals involved in the $\text{Cl}^- \rightarrow \text{H}_2\text{O}$ charge transfer. **(0.2 pt)**

Exercise 4: Decomposing the vibrational frequency shift of CO in the $\text{OBe} \cdots \text{CO}$ complex

In this exercise, we are going to use the adiabatic ALMO-EDA method to investigate the change in the vibrational frequency of CO when it is bound to BeO. All the calculations will be performed at the $\omega\text{B97X-V/def2-TZVPP}$ level of theory.

Let's start with running an "Opt + Freq" calculation for the CO molecule using ω B97X-V/def2-TZVPP. To get a good initial structure, you can draw a CO molecule in IQmol with 3 bonds connecting C and O, and then click "E↓" to minimize the energy. Here I will no longer repeat the procedure of setting up an "Opt + Freq" calculation: you can refer to the detailed procedure in Practice 3. Report the optimized bond length and frequency for "Free CO" in **Table 6**. **(0.25 pt)**

Table 6: Adiabatic ALMO-EDA results for the $\text{OC}\cdots\text{BeO}$ complex at the ω B97X-V/def2-TZVPP level

Stage	CO bond length (Å)	CO frequency (cm^{-1})
Free CO		
FRZ		
POL		
Full		

The input file for the adiabatic EDA calculation has been provided (`OC.BeO.adiabatic_eda_wb97xv.in`), which contains 6 consecutive Q-Chem jobs:

- Jobs 1 & 2: Opt + Freq calculation for the complex on the fully relaxed SCF ("Full") surface
- Jobs 3 & 4: Opt + Freq on the polarized ("POL") surface (with `POL_GEOM = TRUE`)
- Jobs 5 & 6: Opt + Freq on the frozen ("FRZ") surface (with `FRZ_GEOM = TRUE`)

Take a look at the input file and make sure you understand the workflow above, then submit the job to the Expanse server using `submit_qchem -p 16 OC.BeO.adiabatic_eda_wb97xv.in`. It will be submitted to the "shared" queue by default. It may undergo some waiting time in the queue. You can use `squeue -u $USER` to check the job status.

Once finished, open the output using IQmol. You will find three sets of "Frequencies" under the Model View, which, from top to bottom, correspond to the frequencies calculated on the FULL, POL, and FRZ surfaces. Now find the vibrational mode that correspond to the CO stretch (which should not be too far away from the free CO stretch frequency) and report them in the right places in **Table 6**. **(0.3 pt)**

To complete **Table 6**, you now need to fill in the optimized C–O bond length on each of these surfaces. **(0.25 pt)** To find those, you can search for "CONVERGED" in the output (there are 3 occurrences in total), under which you can find the Cartesian coordinates of the optimized structure followed by the **Z-matrix print**. From the Z-matrices you can easily find out the C–O distance in the optimized structures. You may also save the Cartesian coordinates as XYZ files and use IQmol to open them and measure the bond lengths.

Answer the following questions based on the results you obtained in **Table 6**:

1. Is the CO frequency in the fully relaxed complex red- or blue-shifted compared to the free CO stretch frequency? Then describe the respective contributions of frozen interaction, polarization, and charge transfer to the net frequency shift. **(0.3 pt)**
2. Are CO frequencies correlated with the bond lengths at each stage? If so, describe the correlation/trend you observe. **(0.2 pt)**